Lecture 2 (9/8/2025)

$y = \beta_0 + \beta_1 X + u$ ; $u \sim N(0, \sigma^2)$

$\hookrightarrow$ orthogonal to $(1, x)$:

$E(u) = E(ux) = 0$

- Also: $cov(x, u) = E(ux) - E(u)E(x) = 0$
- model assumes independence: $E(u|x) = 0$ (st

Prove:

$$(\beta_0, \beta_1) = \arg\min_b E[(y - \beta_0 - \beta_1 x)^2]$$

$\cdots$ 2

$$\beta_0 = E(y) - \beta_1 E(x)$$
$$= E[x(y - E(y) - \beta_1(x - E(x))] = 0$$
$$= \cancel{E[xy - xE(y) - \beta_1 x - \beta_1 x - \beta_1 E(x)] = 0}$$

$$\Rightarrow \beta_1 = \frac{E[x(y - E(y))]}{E[x - (x - E(x)]}$$

Review:

When do $\tilde{\beta}_1 = \beta_1$

$$\tilde{\beta}_1 = \frac{E[xy]}{E[x^2]} \qquad \beta_1 = \frac{Cov(x, y)}{Var(x)}$$

this happens when $cov(x, y) = E[(x - E(x) \cdot (y - E(y))]$
$= E[xy] - E[x] \cdot E[y]$

# Causal Effects in Policy - Lecture 1

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UW Madison Econ 695

# Course Overview

1. Descriptive modeling
    1. Empirical relationships between $X$ and $Y$: regression analysis
    2. Decomposition techniques
2. Causal modeling: does $X$ "cause" $Y$?
    1. Potential outcomes framework
    2. Randomized control trials: randomly assign treatments
    3. Natural/Quasi experiments: diff-in-diff, instrument variables, LATE, regression discontinuity
3. Statistical Learning (Machine Learning)
    1. Classification methods
    2. Supervised & Unsupervised learning

# 1. Descriptive Modeling

Often we are interested in trying to *summarize the relationship* between some "outcome" $y$ and some other variables $x = (x_1, x_2 ... x_J)$.

- we **aren't** necessarily trying to measure the causal effect of $x_j$ on $y$
- we are trying to take account of the fact that $y$ may be strongly related to some $x_j's$ and only weakly related to others.
- e.g., what is the relationship between earnings ($y$), gender ($x_1$), and other characteristics, like education ($x_2$)?
- our benchmark: the conditional expectation function $E[y|x]$

Benchmark: CEF $E[y|x]$

- CEF itself can be nonlinear.
- we are going to approximate this with a **linear** regression function
    - how does the best linear approximation relate to the CEF?
- we'll consider 2 regression functions in order to get the best linear approx. to CEF:
    - the "population" regression: the function we could estimate with the population ($\infty$ sample)
    - the "sample" regression: the one we can actually estimate with a given sample (drawn randomly from the population?)

# 2. Causal Modeling

- Many debates in economics amount to disputes over the question: does $X$ "cause" $Y$?
- *Examples*
  - if a student were to attend Univ. Minnesota instead of UW, would she get a higher-paid job after college?
  - if an unemployed worker has higher UI benefits, will he/she have longer spell of joblessness?
  - does taking Econ 695 increase the chance of getting a data science job after college?

# 2. Causal Modeling

- *A very empirical notion of causality:* $X$ causes $Y$ if, in an ideal experiment, we could manipulate (randomly change) $X$, leaving other factors ($W$) constant, and observe that the **mean** of the distribution of outcomes of $Y$ has changed.

$$E[Y|X=1,W]-E[Y|X=0,W]$$

- How can we know if the **distribution** of outcomes has changed? We need to be able to see (or at least estimate) two things:
    - the distribution of $Y$ when $X$ is manipulated (or "treated"): $Y|(X=1,W)\sim F_1$
    - the distribution in the absence of manipulation – the *counterfactual* $Y|(X=0,W)\sim F_0$
- Problem: we can't see **both** the outcome under treatment **and** the counterfactual outcome in the absence of treatment
- *How can we resolve the observability problem?* We need a way to infer the counterfactual for the units (people) that are treated

# 2. Causal Modeling

**Possible ideas:**

1. ("observational design"): calculate mean outcomes for people who are treated and those who are not in a given data
    - diff-in-diff, matched control group
2. ("pre-post design"): compare outcomes for people who are treated with their outcomes prior to treatment (i.e., the mean change in outcome after treatment):
    - treatment on the treated (TOT), event study
3. RCT - randomly assign treatment, calculate mean outcomes for T's and C's

4. Instrumental variables: there is some variable $Z$ that shifts $X$ but is unrelated to the unobserved determinants of $Y$:

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + u \\ X &= \pi_0 + \pi_1 Z + \eta \\ Z &\perp u \end{aligned}$$

5. Regression Discontinuity (RD): $X$ shifts discretely ("jumps") when some "running variable" $Z$ passes a threshold. (e.g., eligible for Medicare at age 65, GPA threshold for declaring a major)

# Basic Concepts: Population vs. Sample

- $Y$ is a random variable (r.v.):
  - the randomness comes from the act of random sampling from a population distribution.
  - assume the pop distribution has mean $\mu$, variance $\sigma^2$
- $\{y_1, y_2, ...y_n\}$ is a random sample of $Y$ from the population
- Sample objects:
  - $\overline{y}_n = n^{-1} \sum_{i=1}^{n} y_i$ is the sample mean, which is a "statistic" (a r.v.)
  - $s_n^2 = (n-1)^{-1} \sum_{i=1}^{n} (y_i - \overline{y}_n)^2$ is the sample variance (note $n-1$)
- Properties of the estimators:
  - $E[\overline{y}_n] = \mu$ : the sample mean is an **unbiased** estimator of the pop. mean
  - $Var[\overline{y}_n] = Var[\frac{1}{n} \sum_{i=1}^{n} y_i] = \sigma^2/n$.
  - $E[s_n^2] = \sigma^2$ : the "d.f. corrected" sample var is unbiased for population variance $\sigma^2$ (Prove in class)

# Convergence in Probability & LLN

- *Convergence in Probability*: $Z_1, Z_2, ..$is a sequence of random variables that *converges in probability* to $b$ if for any $\varepsilon > 0$ :

$$lim_{n \to \infty} P(|Z_n - b| < \varepsilon) = 1$$

  Write as $plim Z_n = b$ or $Z_n \to^p b$

- *Weak Law of Large Numbers (WLLN)*. Suppose that $\{y_1, y_2, ..., y_n\}$ is a random sample from a pop with mean $\mu$, variance $\sigma^2$(both finite). Then the sample mean $\bar{y}_n = \frac{1}{n}\sum_{i=1}^{n} y_i$ (a r.v. itself) converges in probability to the population mean

$$plim \overline{y}_n = \mu \text{ or } \overline{y}_n \to^p \mu$$

  in other words, the sample mean is a **consistent** estimator for the population mean.

- *Continuous Mapping Theorem*: Suppose $g$ is a continuous function defined on the same metric space as the random variable $Z_n's$. If $Z_n \to^p b$ and $g$ is continuous at $b$, we have

$$g(Z_n) \to^p g(b)$$

# Weak Law of Large Numbers

To prove WLLN $plim\bar{y}_n = \mu$, we first introduce two important results: Markov inequality; Chebyshev inequality:

1. *Markov*: if
   $X$ is a positively-valued r.v., with $P(X > 0) = 1$, then for any $t > 0$ :

$$P(X \geq t) \leq \frac{E[X]}{t} \tag{1}$$

   Proof:

$$E[X] = \int_0^\infty xf(x)dx = \underbrace{\int_0^t xf(x)dx}_{\geq 0} + \int_t^\infty xf(x)dx$$

$$\geq \int_t^\infty tf(x)dx = t \times Pr(X \geq t)$$

# Weak Law of Large Numbers

2. *Chebychev*: If $X$ is a random variable s.t. $Var[X] \in (0, \infty)$, then for any $t > 0$ :

$$P(|X - E[X]| \geq t) \leq \frac{Var[X]}{t^2} \tag{2}$$

Proof (via Markov): consider r.v. $Y = (X - E[X])^2$. Note $E[Y] = Var[X]$. Using Markov, $\forall\, \tau > 0$,

$$P(Y \geq \tau) \leq \frac{E[Y]}{\tau}$$

$$\text{equiv. to } P(|X - E[X]| \geq \sqrt{\tau}) \leq \frac{E[Y]}{\tau} = \frac{var(X)}{\tau}$$

letting $\tau = t^2$, we have shown (2).

# Weak Law of Large Numbers

- Now we are ready to prove WLLN: suppose the population distribution has finite $(\mu, \sigma^2)$, $plim \overline{y}_n = \mu$. That is,

$$lim_{n\to\infty}(|\overline{y}_n - \mu| < \epsilon) = 1$$

Proof. Applying Chebychev to $\overline{y}_n$: for any $\epsilon > 0$,

$$P(|\overline{y}_n - \mu| \geq \varepsilon) \leq \frac{Var[\overline{y}_n]}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}$$
$$\Rightarrow P(|\overline{y}_n - \mu| < \varepsilon) \geq 1 - \frac{\sigma^2}{n\varepsilon^2}, \text{ where } \frac{\sigma^2}{n\varepsilon^2} \to 0 \text{ as } n \to \infty$$

so $lim_{n\to\infty} P(|\overline{y}_n - \mu| < \varepsilon) = 1$.

- WLLN says that the distribution of the sample mean "collapses" to the point $\mu$ as the sample size gets bigger.

# Convergence in Distribution & Central Limit Theorem

- *Convergence in Distribution*: $Z_1, Z_2, ..$is a sequence of real-valued random variables with cumulative distribution functions $F_1, F_2...$ (cdf $F_i(z) = P(Z_i \leq z)$). It is said to *converge in distribution* to a random variable $Z$ with cdf $F$ if

$$lim_{n\to\infty} F_n(z) = F(z)$$

  for all $z \in \mathbb{R}$ at which $F$ is continuous. Write as $Z_n \to^d Z$.

- The *Central Limit Theorem (CLT)*: Let $\{y_1, ...y_n\}$ be a random sample a population with mean $\mu$, variance $\sigma^2$. The distribution of sample mean $\overline{y}_n$ collapses to a *normal distribution* at the rate $\sqrt{n}$ :

$$lim_{n\to\infty} P(\frac{\sqrt{n}(\overline{y}_n - \mu)}{\sigma} \leq z) = \Phi(z),$$

  where $\Phi$ is cdf for the standard normal $N(0,1)$. Write as $\sqrt{n}(\overline{y}_n - \mu)/\sigma \to^d N(0,1)$.
  - informally, write $\overline{y}_n \approx N(\mu, \sigma^2/n)$

# Convergence in Distribution & Central Limit Theorem

- CLT says that the sample mean is "asymptotically normal", **regardless of the underlying population distribution** (as long as $\mu$ and $\sigma^2$ are finite).
- A key idea of statistics is that for a given $n$ we can step back from the limit and still be "approximately" OK.
  - CLT remains true if, instead of scaling by population $\sigma$, we scale by sample estimate $s_n$:

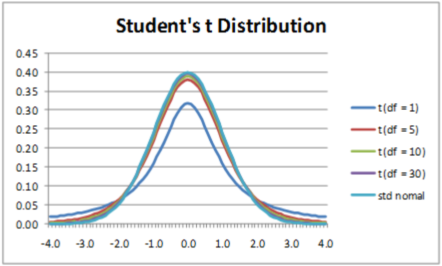
$$\frac{\sqrt{n}(\overline{y}_n - \mu)}{s_n} \text{ approximately } N(0,1)$$

  - The distribution when we use $s_n$instead of $\sigma$(unknown) to scale is known as "t-distribution":

$$\frac{\sqrt{n}(\overline{y}_n - \mu)}{s_n} \sim^d t_{n-1}$$

    where $t_{n-1}$is the t-distribution with $n-1$ degrees of freedom. For large $n$ the $t$ is very close to the standard normal ($t_n \to^d N(0,1)$). For smaller $n$ the $t$ distribution has fatter tails.

# Convergence in Distribution & Central Limit Theorem



# Slutsky Theorem

Slutsky Theorem has 2 parts. We have seen part 1 - contraction mapping.

1. *(Continuous Mapping)*: Suppose $g$ is a continuous function defined on the same metric space as the random variable $Z_n's$. If $Z_n \to^p b$ and $g$ is continuous at $b$, we have

$$g(Z_n) \to^p g(b)$$

2. Given a sequence of random variables $Z_n \to^d N(0, \sigma^2)$ and another sequence of r.v. $A_n \to^p \alpha$, we have

$$Z_n A_n \to^d \alpha \times N(0, \sigma^2) = N(0, \alpha^2 \sigma^2)$$

# Inference - Confidence Interval

*Confidence intervals.*

- Suppose $Z \sim N(0, 1)$. Then we know $Z$ is symmetrically distributed around 0 with a "bell curve" distribution.
- Define $z_p > 0$ as the real number such that $\Phi(z_p) = 1 - p$ (for $p < .5$). This is the point such that $P(Z > z_p) = p$.
- What is the symmetric interval (around 0) such that a standard normal falls in the interval with probability $1 - \alpha$? This is the interval $(-z_{\alpha/2}, z_{\alpha/2})$. Why? Because the probability of falling *above* $z_{\alpha/2}$ is $\alpha/2$, and by symmetry the probability of falling *below* $-z_{\alpha/2}$ is also $\alpha/2$. So with probability of being outside the interval is $\alpha$.

# Inference - Confidence Interval

- For $Z \sim N(0,1)$ $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$. Suppose we have obtained a random sample of some $Y's$ and formed the estimated mean and standard deviation. By the CLT $\sqrt{n}(\overline{y}_n - \mu)/s_n \approx N(0,1)$, so (approximately):

$$P(-z_{\alpha/2} \leq \frac{\sqrt{n}(\overline{y}_n - \mu)}{s_n} \leq z_{\alpha/2}) = 1 - \alpha$$
$$\Rightarrow P(\overline{y}_n - \frac{s_n z_{\alpha/2}}{\sqrt{n}} \leq \mu \leq \overline{y}_n + \frac{s_n z_{\alpha/2}}{\sqrt{n}}) = 1 - \alpha$$

- This is interpreted as: if we kept repeating a sample of size $n$, $(1-\alpha)$ percent of the time the interval $\overline{Y}_n \pm \frac{s_n z_{\alpha/2}}{\sqrt{n}}$ would "capture" the true mean $\mu$. This is called the $(1-\alpha)$ "confidence interval".
- Frequentist view: $\mu$ is a population parameter that is **fixed**. Can we interpret the CI at with with $1-\alpha$ probability $\mu$ takes a value between $\overline{Y}_n \pm \frac{s_n z_{\alpha/2}}{\sqrt{n}}$?

# Univariate Regression

- Consider a normal linear model with a single independent variable:

$$Y = \beta_0 + \beta_1 X + U,\ U \sim N(0, \sigma^2) \tag{3}$$

  - In this model $U \sim N(0, \sigma^2)$ is orthogonal to $(1, X)$: $E[U] = E[UX] = 0$ (note $cov(X, U) = E[UX] - E[U]E[X] = 0$). In fact, the model assumes independence $E[U|X] = 0$, which is stronger than $u_i \perp x_i$!
  - Show (population regression):

$$\begin{aligned} \beta_0 &= E[Y] - \beta_1 E[X] \\ \beta_1 &= \frac{cov(Y, X)}{var(X)} \end{aligned} \tag{4}$$

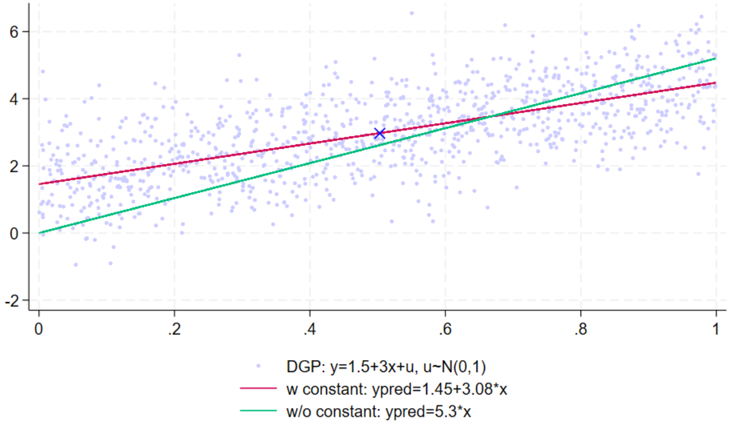
# Univariate Regression

- Given a random sample $\{y_i, x_i\}_{i=1}^n$, we estimate an OLS regression of $y_i$ on 1 and $x_i$:

$$\begin{aligned}(\hat{\beta}_0, \hat{\beta}_1) =& argmin_b \frac{1}{n}\sum_i (y_i - b_0 - b_1 x_i)^2 \\ \rightarrow \hat{\beta}_0 =& \bar{y}_n - \hat{\beta}_1 \bar{x}_n \\ \hat{\beta}_1 =& \frac{\sum_i (y_i - \bar{y})x_i}{\sum_i (x_i - \bar{x})x_i} = \frac{\hat{cov}(x_i, y_i)}{\hat{var}(x_i)}\end{aligned} \tag{5}$$

- Discuss: what's the role of the constant? What if we a run a regression without a constant?

# Univariate Regression



# Inference - Consistency

We want to do inference on $(\hat{\beta}_0, \hat{\beta}_1)$: how are the estimates compared to the true parameter $(\beta_0, \beta_1)$?

Plug $y_i = \beta_0 + \beta_1 x_i + u_i$ into $\hat{\beta}_1$ and let $\bar{u} := \bar{y} - \beta_0 - \beta_1 \bar{x}$:

residuals

$$\hat{\beta}_1 = \frac{\sum_i (y_i - \bar{y})x_i}{\sum_i (x_i - \bar{x})x_i} = \frac{\sum_i (\beta_0 + \beta_1 x_i + u_i - \beta_0 - \beta_1 \bar{x} - \bar{u})x_i}{\sum_i (x_i - \bar{x})x_i}$$

we can use CLT to solve

$$= \beta_1 + \frac{\sum_i x_i (u_i - \bar{u})}{\sum_i (x_i - \bar{x})x_i}$$

- First, show consistency: $\hat{\beta}_0 \to^p \beta_0$, $\hat{\beta}_1 \to^p \beta_1$. By Weak LLN,

they asymptotically converge to their population parameter

$$\frac{1}{n-1} \sum_i x_i (u_i - \bar{u}) \to^p cov(X, U) = 0$$

$$\frac{1}{n-1} \sum_i (x_i - \bar{x})x_i \to^p var(X)$$

by CMP, $\hat{\beta}_1 = \beta_1 + \dfrac{\frac{1}{n-1}\sum_i x_i (u_i - \bar{u})}{\frac{1}{n-1}\sum_i (x_i - \bar{x})x_i} \to^p \beta_1 + \dfrac{cov(X, U)}{var(X)} = \beta_1$

continuous mapping theorem

and $\hat{\beta}_0 \to^p E[Y] - \beta_1 E[X] = \beta_0$

# Inference - Asymptotic Distribution

Second, derive the asymptotic distribution of $\hat{\beta}_1$ and find the 95% confidence interval for $\beta_1$.

$\sqrt{n}(\hat{\beta}_1 - \beta_0) = \dfrac{\frac{1}{\sqrt{n}} \sum_i x_i (u_i - \bar{u})}{\frac{1}{n} \sum_i (x_i - \bar{x})^2}$

$Var(x_i u_i) = \underbrace{E[(x_i u_i)(x_i u_i)]}_{= E[x^2 u^2]} - \underbrace{E[xu]^2}_{=0}$

Note for large sample, $n - 1 \approx n$.

- By the Central Limit ThM,

$$\frac{1}{\sqrt{n}} \sum_i x_i (u_i - \bar{u}) = \frac{1}{\sqrt{n}} \sum_i (x_i - \bar{x})(u_i - \bar{u})$$

$$\to^d N(0, E[(X - E[X])U^2]) \underbrace{= N(0, \sigma^2 var(X)))}_{(*)}$$

(*) comes from the law of iterated expectations and $\underbrace{E[U^2|X] = \sigma^2}_{\text{generally not true}}$ under the assumption that $U \sim N(0, \sigma^2)$.

  - By WLLN, the denominator $\frac{1}{n-1} \sum_i (x_i - \bar{x}) x_i \to^p var(X)$

- We can apply Slutsky Theorem:

$$\sqrt{n}(\hat{\beta}_1 - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_i x_i (u_i - \bar{u})}{\hat{var}(X)} \to^d N(0, \frac{\sigma^2}{var(X)})$$

# OLS in Vector Notation

Suppose we are interested in the OLS regression model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \tag{6}$$

where $i = 1 \ldots N$ indexes elements of a sample. Here $(x_{1i}, x_{2i}, y_i)$ are observed values of two *covariates* and our *outcome of interest* $(y)$ for unit $i$. We can define the 3-row vectors $x_i$ and $\beta$ :

$$x_i = \begin{pmatrix} 1 \\ x_{1i} \\ x_{2i} \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}$$

Using these vectors we can write the model in vector notation:

$$y_i = x_i' \beta + u_i \tag{7}$$

and the OLS estimator:

$$\hat{\beta} = argmin_b \sum_i (y_i - x_i' b)^2$$

Differentiate the dot product $x_i' b$ w.r.t. $b$? $\boxed{\frac{\partial x_i' b}{\partial b} = x}$ b/c $\begin{bmatrix} 1 & x_{1i} & x_{2i} \end{bmatrix} \cdot \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$

$x\beta$

# OLS in Vector Notation

$$\frac{\partial(x_i' b)}{\partial b} = \begin{bmatrix} \frac{\partial(x_i' b)}{\partial b_1} \\ \frac{\partial(x_i' \beta)}{\partial b_2} \\ \cdots \\ \frac{\partial(x_i' \beta)}{\partial b_K} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \cdots \\ x_{iK} \end{bmatrix} = x_i$$

So

$$\frac{\partial \sum_i (y_i - x_i' b)^2}{\partial b} = -2 \sum_i x_i \left(y_i - x_i' b\right) = 0$$

$$\Rightarrow \hat{\beta} = \left( \sum_i x_i x_i' \right)^{-1} \left( \sum_i x_i y_i \right)$$

# OLS in Vector Notation

$$\frac{\partial(x_i' b)}{\partial b} = \begin{bmatrix} \frac{\partial(x_i' b)}{\partial b_1} \\ \frac{\partial(x_i' \beta)}{\partial b_2} \\ \cdots \\ \frac{\partial(x_i' \beta)}{\partial b_K} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \cdots \\ x_{iK} \end{bmatrix} = x_i$$

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# OLS in Vector Notation

$$\frac{\partial(x_i'b)}{\partial b} = \begin{bmatrix} \frac{\partial(x_i'b)}{\partial b_1} \\ \frac{\partial(x_i'\beta)}{\partial b_2} \\ \dots \\ \frac{\partial(x_i'\beta)}{\partial b_K} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \dots \\ x_{iK} \end{bmatrix} = x_i$$

So

$$\frac{\partial \sum_i (y_i - x_i'b)^2}{\partial b} = -2\sum_i x_i \left(y_i - x_i'b\right) = 0$$

$$\Rightarrow \hat{\beta} = \left(\sum_i x_i x_i'\right)^{-1} \left(\sum_i x_i y_i\right)$$

# Causal Effects in Policy - Lecture 2

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# This lecture

- Prove the Law of Iterated Expectations (LIE)
- Conditional Expectation Function (CEF) - Definition & 3 Properties
- Best Linear Predictor as an approximation to CEF
- Regressions with Discrete Covariates (Dummies)

# Law of Iterated Expectations

- $(X, Y)$ are two r.v.'s with joint pdf $f(X, Y)$
  - Marginal densities: $f(X) = \int_Y f(X, Y)dY$, and conditional density: $f(Y|X) = \frac{f(X,Y)}{f(X)}$.
- Expectation: $E[Y] = \int_Y f(Y)dY$, conditional $E[Y|X] = \int_Y Yf(Y|X)dY$, which is a function of $X$.
- Law of iterated expectations (LIE):

$$E[Y] = E[E[Y|X]] \tag{1}$$

write $E(y)$ as an iterated expectation

$E(y) = \int y f(y)\, dy = E_x[E[y|x]]$

$\Rightarrow E(y|x) = \int_y f(y|x)dy$

$\Rightarrow E_x[E[y|x]] = \int_x E[y|x] f(x)dy$

$E_x[g(x)] = \int_x \int_y y f(y|x)\, dy \cdot f(x)\, dx$

Proof (continuous r.v.'s):

$$\begin{aligned} E[E[Y|X]] = \int_X E[Y|X]f(X)dx &= \int_X \left( \int_Y Yf(Y|X)dY \right) f(X)dX \\ &= \int_Y Y \left( \int_X f(Y|X)\, f(X)dX \right) dY \\ &= \int_Y Yf(Y)\, dY = E[Y] \end{aligned}$$

# Conditional Expectation Function

- CEF $E[Y|X]$ is a function of $X$ that represents the expectation of $Y$ conditional on random variables $X$.
- Let $\epsilon = Y - E[Y|X]$. *Two excellent properties of CEF:*
  1. $E[\epsilon] = 0$, $E[\epsilon|X] = 0$ and $E[\epsilon\, h(X)] = 0$ for *any* function of $X$.
     Proof:

$$E[\epsilon] = E[Y] - E[E[Y|X]] = 0 \text{ by LIE} \tag{2}$$
$$E[\epsilon|X] = E[Y|X] - E[E[Y|X]|X] = E[Y|X] - E[Y|X] = 0$$

*Remark:* $E[\epsilon|X] = 0$ means $\epsilon$ is mean independent of $X$. With $E[\epsilon|X] = 0$, by LIE,

$$E[\epsilon h(X)] = E[E[\epsilon h(X)|X]] = E[E[\epsilon|X]\, h(X)] = 0 \tag{3}$$

## Proof of:

$$\varepsilon_i = Y - E[Y|X]$$

$$\Rightarrow E[\varepsilon] = E[Y - E[Y|X]]$$
$$= E[Y] - \underbrace{E[E[Y|X]]}_{= E[x]}$$

$$\Rightarrow E[\varepsilon|x] = E[Y|x] - \underbrace{E[E[Y|x]|x]}_{= E[Y|x]}$$

$$= 0$$ $\Rightarrow$ mean independence??

$$E[\varepsilon \cdot h(x)] = E_x[E[\varepsilon \cdot h(x)|x]]$$
$$= E[E[\varepsilon|x]h(x)]$$
$$= 0$$

# Conditional Expectation Function

2. the function $E[Y|X]$ minimizes $E[(Y - m(X))^2]$
Proof: given any function $m$ of r.v. X,

$$
\begin{aligned}
Y - m(X) &= Y - E[Y|X] + E[Y|X] - m(X) \\
\Rightarrow (Y - m(X))^2 &= (Y - E[Y|X])^2 + (E[Y|X] - m(X))^2 \\
&\quad +2\,(Y - E[Y|X])\underbrace{(E[Y|X] - m(X))}_{\text{funx of X: } h} \\
\Rightarrow E[(Y - m(X))^2] &= E[\epsilon^2] + \underbrace{E[(E[Y|X] - m(X))^2]}_{\geq 0} + \underbrace{2E[\epsilon h(X)]}_{=0}
\end{aligned}
$$

The minimizing choice is $m(X) = E[Y|X]$!
So: we've established that among the functions of $X$, the CEF $E[Y|X]$ gives the "best guess" for $Y$ in the sense of minimizing $E[(Y - m(X))^2]$.

# Linear Approximation to CEF

- *Problem: we don't know the conditional density $f(Y|X)$.*
- Solution: we'll use the "linear regression function": combination $x_i\beta$.
- Recall: given a sample of size $N$ the OLS regression coefficients $\beta$ solve:

$$\min_{\beta} \sum_{i=1}^{N} (y_i - x_i'\beta)^2$$

- Consider WLLN for the r.v. $(y_i - x_i'\beta)^2$ : we have $\frac{1}{N}\sum_{i=1}^{N}(y_i - x_i'\beta)^2 \to^p E[(y_i - x_i'\beta)^2]$
- The "infeasible" (or population) OLS estimator solves:

$$\min_{\beta} E[(y_i - x_i'\beta)^2] \tag{4}$$

$$\Rightarrow \beta^* = E[x_i x_i]^{-1} E[x_i y_i]$$

This gives us the best linear predictor of Y given X: $E^*[y_i|x_i] = x_i'\beta^*$ (proof below).

# Linear Approximation to CEF

How does $x_i'\beta^*$ relate to $E[y_i|x_i]$?

1. If the CEF is linear $E[y_i|x_i] = x_i'\beta^e$, then $\beta^* = \beta^e$.
   Why? Recall that if we define the CEF error $\varepsilon_i = y_i - E[y_i|x_i]$,

   $$E[x_i\varepsilon_i] = 0 \Rightarrow E[x_i(y_i - x_i'\beta^e)] = 0 \Rightarrow \beta^e = \beta^*$$

   This means that *if the true CEF is linear*, then the infeasible OLS represents the CEF.

   - This happens when $x's$ are dummies since $E[y_i|x_i]$ is $E[y_i|i$ in group k]

# Linear Approximation to CEF

2. $E^*[y_i|x_i] = x_i'\beta^*$ is the "best" linear approx. to $E[y_i|x_i]$ (best as in *minimum-MSE*)

$$\beta^* = argmin_\beta E[(E[y_i|x_i] - x_i'\beta)^2] \tag{5}$$

Proof:

$$\begin{aligned} y_i - x_i'\beta &= y_i - E[y_i|x_i] + E[y_i|x_i] - x_i'\beta \\ \Rightarrow E[(y_i - x_i'\beta)^2] &= E[\varepsilon_i^2] + E[(E[y_i|x_i] - x_i'\beta))^2] \\ &\quad +2\underbrace{E[\varepsilon_i\,(E[y_i|x_i] - x_i'\beta)]}_{=0} \\ E[(E[y_i|x_i] - x_i'\beta))^2] &= E[(y_i - x_i'\beta)^2] - E[\epsilon_i^2] \end{aligned}$$

So $x_i'\beta^*$ that minimizes the mean squared error also minimizes $E[(E[y_i|x_i] - x_i'\beta))^2]$.

We can think of the "population regression" as:

$$y_i = x_i'\beta^* + u_i$$

which satisfies $E[x_i u_i] = 0$ (why?). Unless the CEF is linear, we won't have $E[u_i|x_i] = 0$.

# Feasible OLS

- We have established that $E^*[y_i|x_i] = x_i'\beta^*$ is the best linear approximation to the CEF $E[y_i|x_i]$.

$$\begin{aligned} \beta^* &= argmin_\beta \, E[(y_i - x_i'\beta)^2] \\ &= E[x_i x_i]^{-1} E[x_i y_i] \end{aligned}$$

- How do find an estimate for $\beta^*$? The feasible OLS minimizes the sum of squared residuals (SSR) in the sample:

$$\hat{\beta} = argmin_\beta \sum_{i=1}^{N} (y_i - x_i'\beta)^2 \tag{6}$$

The FOC (in vector form) are:

$$\sum_{i=1}^{N} -2x_i(y_i - x_i'\beta) = 0$$

# Feasible OLS

which implies that

$$\begin{aligned}\sum_{i=1}^{N} x_i x_i' \beta &= \sum_{i=1}^{N} x_i y_i \\ \Rightarrow \hat{\beta} &= [\sum_{i=1}^{N} x_i x_i']^{-1} \sum_{i=1}^{N} x_i y_i \\ &= [\frac{1}{N}\sum_{i=1}^{N} x_i x_i']^{-1} \left(\frac{1}{N}\sum_{i=1}^{N} x_i y_i\right)\end{aligned}$$

Now plug in the population regression: $y_i = x_i'\beta^* + u_i$, where $E[u_i x_i] = 0$,

$$\begin{aligned}\hat{\beta} &= [\frac{1}{N}\sum_{i=1}^{N} x_i x_i']^{-1} \frac{1}{N}\sum_{i=1}^{N} x_i(x_i'\beta^* + u_i) \\ &= \beta^* + [\frac{1}{N}\sum_{i=1}^{N} x_i x_i']^{-1} [\frac{1}{N}\sum_{i=1}^{N} x_i u_i] \qquad (7)\end{aligned}$$

# Feasible OLS

- Consistency: $\hat{\beta} \to^p \beta^*$ by WLLN and Slutsky Theorem.
- Asymptotic distribution:
  - by the Central Limit ThM and $E[x_i u_i] = 0$:

$$\sqrt{N}\left(\frac{1}{N}\sum_{i=1}^{N} x_i u_i - E[x_i u_i]\right) \to^d N(0, E[x_i x_i' u_i^2])$$

note the variance of $K$-by-1 vector $x_i u_i$ ($u_i$ is a scalar): $var(x_i u_i) = E[x_i u_i u_i x_i'] - E[x_i u_i]E[x_i u_i]' = E[x_i x_i' u_i^2] - 0_{K \times K} = E[x_i x_i' u_i^2]$.

  - By Slutsky ThM, we have:

$$\sqrt{N}(\hat{\beta} - \beta^*) \to^d N(0, E[x_i x_i']^{-1} E[x_i x_i' u_i^2] E[x_i x_i']^{-1}) \qquad (8)$$

when do we have $E[x_i x_i' u_i^2] = E[x_i x_i'] \times var(u_i)$? Apply LIE:

$$\begin{aligned} E[x_i x_i' u_i^2] &= E_x[E[x_i x_i' u_i^2 | x]] \\ &= E_x[x_i x_i' E[u_i^2 | x]] \end{aligned}$$

Homoskedasticity: $E[u_i^2|x] = E[u_i^2]$. Otherwise the variance of residual vary by $x$. We need to compute robust heteroskedasticity-robust standard errors.

# When is CEF linear? $E[y|x] = E^*[y|x]$

- Recall in the linear normal model, we assume:

$$y_i = \beta_0 + \beta_1 x_i + u_i,\ u_i \sim N(0, \sigma^2) \tag{9}$$

in which the normally distributed $U$ is independent of $X$: $E[u_i|x_i] = 0$. This is a special case where the CEF is linear:

$$E[y_i|1, x_i] = \beta_0 + \beta_1 x_i$$

note the constant 1 is often merged into $x_i$:

$$E[y_i|x_i] = x_i'\beta = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

- Another import case: regressions with discrete regressors! $x_i$ are indicators/dummies...

# Regressions with Dummy Variables

- Consider a (population) regression with a binary independent variable $D_i \in \{0,1\}$:

$$\begin{aligned} y_i &= \alpha + \beta_1 D_i + \epsilon_i \\ E[\epsilon_i] &= E[\epsilon_i D_i] = 0 \end{aligned} \tag{10}$$

Recall this represents the best linear projection of $y$ on $D$, $E^*[y_i|D_i] = \alpha + \beta_1 D_i$.

$$\begin{aligned} (\alpha, \beta_1) &= argmin_{(a,b)} E[(y_i - a - bD_i)^2] \\ \alpha &= E[y_i] - \beta_1 E[D_i] \\ 0 &= E[(y_i - \alpha - \beta_1 D_i) D_i] = E[\epsilon_i D_i] \\ \rightarrow \beta_1 &= \frac{cov(y_i, D_i)}{var(D_i)} \text{ shown in lecture 1 (univariate reg)} \end{aligned}$$

# Regressions with Dummy Variables

- Since $D_i$ is a dummy variable/indicator, we have $E[D_i^2] = E[D_i] = Pr(D_i = 1)$. Denote by $p = E[D_i]$.

$$\begin{aligned} E[y_i D_i] =& E_{D_i}[E[y_i D_i | D_i]] \\ =& p \times E[y_i D_i | D_i = 1] + (1-p) \times \underbrace{E[y_i D_i | D_i = 0]}_{0} \\ =& p \times E[y_i | D_i = 1] \end{aligned}$$

therefore,

$$\begin{aligned} cov(y_i, D_i) &= E[y_i D_i] - E[y_i]E[D_i] = E[y_i | D_i = 1] - E[y_i] * p \\ var(D_i) &= E[D_i^2] - E[D_i]^2 = p(1-p) \end{aligned}$$

- denote by ,prove:

$$\begin{aligned} \beta_1 &= E[y_i | D_i = 1] - E[y_i | D_i = 0] \\ \alpha &= E[y_i | D_i = 0] \end{aligned} \tag{11}$$

# Regressions with Dummy Variables

- In summary, with a single binary variable, the best linear projection of $y_i$ on $(1, D_i)$ is:

$$\begin{aligned} E^*[y_i|D_i] &= E[y_i|D_i = 0] + D_i \times (E[y_i|D_i = 1] - E[y_i|D_i = 0]) \\ &= E[y_i|D_i] \text{ CEF!!} \end{aligned}$$

  we have shown that in this case the conditional expectation of any outcome $y_i$ on a discrete variable is Linear! The constant is the mean of the outcome among $D_i = 0$, and the slope is the difference in means for the two groups.

# Regressions with Dummy Variables

- In a randomized control trial, individuals are In the RCT we have 2 groups: the treatment group, with $D_i = 1$, and the control group, with $D_i = 0$. So we could fit a regression model to the "pooled" data

$$y_i = \alpha + \beta D_i + \epsilon_i$$

The treatment effect of interest is $E[y_i|D_i = 1] - E[y_i|D_i = 0]$. (We will discuss this in the potential outcome framework.)

- The (feasible) OLS estimator: let $\bar{Y}_0$ denote the sample mean of the outcome for the control group, and $\bar{Y}_1$ for the treatment group,

$$\begin{aligned}\hat{\alpha} &= \overline{Y}_0 \;\; \rightarrow^P E[y_i|D_i = 0] \\ \hat{\beta} &= \overline{Y}_1 - \overline{Y}_0 \;\; \rightarrow^p E[y_i|D_i = 1] - E[y_i|D_i = 0]\end{aligned} \tag{12}$$

So we get the estimated mean of the control group in the intercept, and the estimated treatment effect in the coefficient of $D_i$.

# Regressions with Multiple Dummies

- Now we generalize the result to a regression with multiple indicators/dummies.
- Suppose individuals belong to $G$ mutually exclusive groups. For example, education level: less than HS, HS, College and above.
- Define $D_{gi} = 1[i \in g]$ for $g = 1, 2, ..., G$. Because of the mutual exclusivity, $D_{gi} \times D_{ki} = 0$ for any $g \neq k$.
- We posit a population regression:

$$y_i = x_i' \beta + u_i \tag{13}$$

$$x_i := \begin{bmatrix} D_{1i} \\ ... \\ D_{Gi} \end{bmatrix}$$

Given a random sample, let $N_g = \sum_i D_{gi}$ denote the num. individuals in group $g$. The fraction of individuals in $g$: $\bar{p}_g = \frac{1}{N} \sum_i D_{gi} = \frac{N_g}{N}$. Recall the OLS estimator:

$$\hat{\beta} = [\frac{1}{N} \sum_i x_i x_i']^{-1} \frac{1}{N} \sum_i x_i y_i$$

# Regressions with Multiple Dummies

- Now let's break it down:

$$\frac{1}{N}\sum_i x_i x_i' = \begin{pmatrix} \frac{1}{N}\sum D_{1i}^2 & \frac{1}{N}\sum D_{1i}D_{2i} & \dots & \frac{1}{N}\sum D_{1i}D_{Gi} \\ \frac{1}{N}\sum D_{2i}D_{1i} & \frac{1}{N}\sum D_{2i}^2 & \dots & \\ \dots & & & \\ & & & \frac{1}{N}\sum D_{Gi}^2 \end{pmatrix}$$

$$= \begin{pmatrix} \bar{p}_1 & 0 & \dots & 0 \\ 0 & \bar{p}_2 & \dots & 0 \\ \dots & & & \\ 0 & & & \bar{p}_G \end{pmatrix}$$

And with

$\frac{1}{N}\sum_{i=1}^{N} D_{gi}y_i = \frac{1}{N}\sum_{i\in g} y_i = \frac{N_g}{N} \times \left(\frac{1}{N_g}\sum_{i\in g} y_i\right) = \bar{p}_g \times \bar{y}_g,$

$$\frac{1}{N}\sum_i x_i y_i = \begin{pmatrix} \frac{1}{N}\sum D_{1i}y_i \\ \frac{1}{N}\sum D_{2i}y_i \\ \dots \\ \frac{1}{N}\sum D_{Gi}y_i \end{pmatrix} = \begin{pmatrix} \bar{p}_1\bar{y}_1 \\ \bar{p}_2\bar{y}_2 \\ .. \\ \bar{p}_G\bar{y}_G \end{pmatrix}$$

# Regressions with Multiple Dummies

So

$$\hat{\beta} = \begin{pmatrix} \overline{p}_1 & 0 & \dots & 0 \\ 0 & \overline{p}_2 & \dots & 0 \\ \dots & & & \\ 0 & & & \overline{p}_G \end{pmatrix}^{-1} \begin{pmatrix} \overline{p}_1 \overline{y}_1 \\ \overline{p}_2 \overline{y}_2 \\ .. \\ \overline{p}_G \overline{y}_G \end{pmatrix} = \begin{pmatrix} \overline{y}_1 \\ \overline{y}_2 \\ .. \\ \overline{y}_G \end{pmatrix} \tag{14}$$

Also:

$$\overline{x} = \frac{1}{N} \sum_i x_i = \begin{pmatrix} \frac{1}{N} \sum D_{1i} \\ \dots \\ \frac{1}{N} \sum D_{Gi} \end{pmatrix} = \begin{pmatrix} \overline{p}_1 \\ .. \\ \overline{p}_G \end{pmatrix}$$

So obviously

$$\overline{y} = \overline{x}' \hat{\beta} = \sum_g \overline{p}_g \overline{y}_g \tag{15}$$

So the OLS regression is just a way to get the group-specific means.

# Regressions with Multiple Dummies

The population version:

$$
\begin{aligned}
E^*[y_i|x_i] &= E[x_i]'\beta \\
&= \sum_g E[D_{gi}] \times E[y_i|D_{gi} = 1] \\
&= E[y_i|x_i]
\end{aligned}
$$

the linear projection is also the CEF!

# Regressions with Multiple Dummies

- What if there is a constant in $x_i$: $x_i = [1, D_{1i}, D_{2i}, ..., D_{Gi}]$? $\sum_{g=1}^{G} D_{gi} = 1$ (the first element of $x_i$)
  - Multicollinearity: The matrix $\frac{1}{N}\sum_i x_i x_i'$ is singular (not invertible)! the sum of columns 2-$(G+1)$ equals the first column (1's).

$$\frac{1}{N}\sum_i x_i x_i' = \begin{pmatrix} 1 & \bar{p}_1 & 0 & ... & 0 \\ 1 & 0 & \bar{p}_2 & ... & 0 \\ ... & ... & ... & ... & \\ 1 & 0 & 0 & & \bar{p}_G \end{pmatrix}$$

that is, the covariates $x_i = [1, D_{1i}, D_{2i}, ..., D_{Gi}]$ with dummies for every possible group and a constant are linearly dependent!

# Regressions with Multiple Dummies

- In Stata (see reg_dummy.do), you may run "reg y ibn.group" which automatically generate $D_{1i}, D_{2i}, ..., D_{Gi}$ for you. By default it will keep constant, but drop a group as base to ensure $\frac{1}{N}\sum_i x_i x_i'$ is nonsingular:

```
. reg tempjuly ibn.region, robust  // West as base
note: 4.region omitted because of collinearity.

Linear regression                               Number of obs     =        954
                                                F(3, 950)         =     392.53
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4247
                                                Root MSE          =     4.1746

------------------------------------------------------------------------------
             |               Robust
    tempjuly | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      region |
         NE  |   1.241406   .4451532     2.79   0.005     .3678092    2.115004
     N Cntrl |    1.35866   .4451531     3.05   0.002     .4850625    2.232257
      South  |   8.881006   .4468209    19.88   0.000     8.004136    9.757876
       West  |          0  (omitted)
             |
       _cons |   72.10859    .405254   177.93   0.000      71.3133    72.90389
------------------------------------------------------------------------------
```

# Regressions with Multiple Dummies

- And if you suppress constant, Stata will keep all groups and the coef on $D_{gi}$: $\hat{\beta}_g = \bar{y}_g$,

```
. reg tempjuly ibn.region, robust nocons

Linear regression                               Number of obs     =        954
                                                F(4, 950)         >   99999.00
                                                Prob > F          =     0.0000
                                                R-squared         =     0.9969
                                                Root MSE          =     4.1746
```

| tempjuly | Coefficient | Robust std. err. | t | P>\|t\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| region | | | | | | |
| NE | 73.35 | .1842025 | 398.20 | 0.000 | 72.98851 | 73.71149 |
| N Cntrl | 73.46725 | .1842024 | 398.84 | 0.000 | 73.10576 | 73.82874 |
| South | 80.9896 | .1881971 | 430.34 | 0.000 | 80.62027 | 81.35893 |
| West | 72.10859 | .405254 | 177.93 | 0.000 | 71.3133 | 72.90389 |

# Regressions with Multiple Dummies

- In Python (see reg_dummy.py), I use statsmodel-OLS. By default it will keep constant, but drop a group as base to ensure $\frac{1}{N}\sum_i x_i x_i'$ is nonsingular: to be consistent with Stata, I use “West” as the base/reference group.

```
==============================================================================
Dep. Variable:                tempjuly   R-squared:                       0.425
Model:                             OLS   Adj. R-squared:                  0.423
Method:                  Least Squares   F-statistic:                     392.5
Date:                 Thu, 04 Sep 2025   Prob (F-statistic):          8.61e-166
Time:                         10:21:30   Log-Likelihood:                -2715.0
No. Observations:                  954   AIC:                             5438.
Df Residuals:                      950   BIC:                             5457.
Df Model:                            3
Covariance Type:                   HC1
===============================================================================================================
                                                coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------------------------------------
Intercept                                    72.1086      0.405    177.934      0.000      71.314      72.903
C(region, Treatment(reference='West'))[T.NE]  1.2414      0.445      2.789      0.005       0.369       2.114
C(region, Treatment(reference='West'))[T.N Cntrl] 1.3587  0.445      3.052      0.002       0.486       2.231
C(region, Treatment(reference='West'))[T.South] 8.8810    0.447     19.876      0.000       8.005       9.757
==============================================================================
```

# Regressions with Multiple Dummies

- Suppress the constant by adding "0+" before other covariates. Python - smf.ols also keeps all groups and the coef on $D_{gi}$: $\hat{\beta}_g = \bar{y}_g$,

```
==============================================================================
Dep. Variable:               tempjuly   R-squared:                       0.425
Model:                            OLS   Adj. R-squared:                  0.423
Method:                 Least Squares   F-statistic:                       nan
Date:                Thu, 04 Sep 2025   Prob (F-statistic):                nan
Time:                        10:21:30   Log-Likelihood:                -2715.0
No. Observations:                 954   AIC:                             5438.
Df Residuals:                     950   BIC:                             5457.
Df Model:                           3
Covariance Type:                  HC1
=====================================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------
C(region)[NE]        73.3500      0.184    398.203      0.000      72.989      73.711
C(region)[N Cntrl]   73.4673      0.184    398.840      0.000      73.106      73.828
C(region)[South]     80.9896      0.188    430.344      0.000      80.621      81.358
C(region)[West]      72.1086      0.405    177.934      0.000      71.314      72.903
```

# OLS in Vector Notation

$$\frac{\partial(x_i'b)}{\partial b} = \begin{bmatrix} \frac{\partial(x_i'b)}{\partial b_1} \\ \frac{\partial(x_i'\beta)}{\partial b_2} \\ \cdots \\ \frac{\partial(x_i'\beta)}{\partial b_K} \end{bmatrix} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \cdots \\ x_{iK} \end{bmatrix} = x_i$$

So

$$\frac{\partial \sum_i (y_i - x_i'b)^2}{\partial b} = -2\sum_i x_i\left(y_i - x_i'b\right) = 0$$

$$\Rightarrow \hat{\beta} = \left(\sum_i x_i x_i'\right)^{-1}\left(\sum_i x_i y_i\right)$$